

COMP1017 Test 1 Answers

1. possible solutions are

$$(p \wedge q) \wedge \neg r \quad \text{or} \\ \neg((p \wedge q) \rightarrow r)$$

2. (a)

$P(x)$: x is personal data

$I(x)$: x is information

$C(x)$: x is in respect of commercial transaction

$D(x)$: x is data subject

$F(x, y)$: x is identified in y

$T(x, y)$: x is identifiable in y

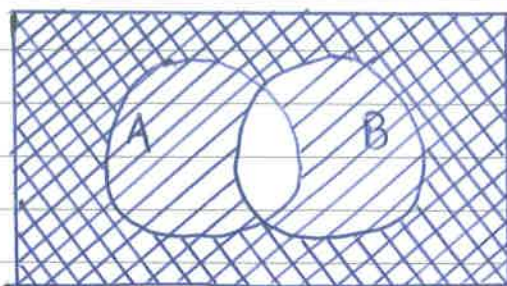
$Q(x, y, z)$: x is combined from y and z

(b)

$$\forall x P(x) \leftrightarrow I(x) \wedge C(x) \wedge (\exists y, F(y, x) \vee (\exists z, s, \\ I(z) \wedge I(s) \wedge Q(z, x, s) \wedge F(y, z)))$$

some may argue that the last predicate should be $T(y, z)$. However, if you read carefully the meaning of the sentence, "identifiable" becomes "identified" if information are combined.

3. (a)



Initially,

 area represents $\overline{A \cap B}$

 area represents $\overline{A \cup B}$

After intersection  area being taken out of $\overline{A \cap B}$,

the area of  is the end result, which is $(\overline{A \cup B}) - (A \cap B)$

3(b) for the operator Δ .

Domain : $P(S) \times P(S)$ where $P(S)$ is the powerset of S

$$|P(S) \times P(S)| = 2^{10} \times 2^{10} = 2^{20}$$

codomain: $P(S)$ $|P(S)| = 2^{10}$

(c) $f \circ \Delta$ does not exist because the codomain of Δ is $P(S)$ whilst the domain of f is $P(S) \times P(S)$.

$$\Delta \circ f(A, B) = \Delta(f(A, B))$$

$$= \Delta(f(\{2, 4, 5, 6\}, \{1, 2, 3, 4, 6, 7\}))$$

$$= \Delta(\{1, 2\}, \emptyset)$$

$$= \{1, 2, \dots, 10\} - \{3, 4, \dots, 10\} = \{1, 2\}$$